Nathaniel Johnson
INFO-C 450: System Design
Professor Jabin
04/17/2026

Module 12: Integrated Project

Part 1: Project Overview and Business Case

1.1 Company Description

- For this project the fictional company I chose is the Cinema Flow theater chain I have been working on. This fictional company is a small independent movie theater chain that operates in many locations. Cinema Flow currently relies on a manual system for managing movies, ticket bookings and customer records.

1.2 Problem Statement

- The current system Cinema Flow relies heavily on manual ticket sales at the counter, phone-based reservations, spreadsheet-based tracking of showtimes and seat availability. This can cause issues such as double booking, long wait times, risk of human error and no real-time seat availability.

1.3 Proposed System

- The proposed solution is a web-based movie ticket booking system that would allow customers to view movies and showtimes online, real-time seat selection, booking and cancellation of tickets.

1.4 Objectives of the System

- The main objective of the system is to improve customer experience through online booking, provide real-time seat availability, reduce booking errors and manual work, and allow administrators to manage movie listings and showtimes.

1.5 Feasibility Analysis

- **Technical Feasibility:** The system is technically feasible because it can be developed using existing technologies such as Flask and MySQL.
- **Economic Feasibility:** This system is economically feasible because it development costs are low, reduce staffing needs for manual ticket handling, increase revenue, and minimize maintenance costs.
- **Operational Feasibility:** The system is operationally feasible because the staff can easily be trained on it, customers are familiar with online booking systems, and the system improves current workflows rather than replacing them.

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Part 2: Project Management

2.1 Project Overview

- The Cinema Flow system will be developed over a structured timeline using a phased approach. The project includes planning, analysis, design, development, testing, and deployment stages. The project will follow a structured SDLC (Systems Development Life Cycle) approach.

2.2 Gantt Chart

Task Phase	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**
Project Planning	✅					
Requirements Gathering	✅	✅				
System Design		✅	✅			
Development			✅	✅		
Testing				✅	✅	
Deployment & Training					✅	✅

2.3 Major Milestones

- Milestone 1: Project Approval
- Milestone 2: Requirements Finalized
- Milestone 3: System Design Completed
- Milestone 4: Development Completed
- Milestone 5: Testing Completed
- Milestone 6: System Deployment

2.4 Risk Analysis and Mitigation Plan

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Risk	**Impact**	**Likelihood**	**Mitigation Strategy**
Requirement misunderstandings	High	Medium	Conduct regular stakeholder meetings
Technical issues (bugs/errors)	High	Medium	Perform continuous testing during development
Database failures	High	Low	Implement backups and recovery plans
User resistance to new system	Medium	Medium	Provide training and user support
Project delays	Medium	Medium	Use clear scheduling and milestone tracking

Part 3 Requirements Modeling

3.1 Functional Requirements

- The system must allow customers to view available movies and showtimes.
- The system must allow customers to select seats and book tickets.
- The system must allow administrators to add and manage movies and showtimes.

3.2 Non-Functional Requirements

- The system should be easy to learn and use for both customers and theater staff.
- The system should load movie listings, showtimes, and seat maps quickly to avoid delays.

3.3 Gathering Requirements

- Requirements for the Cinema Flow system would be gathered using several techniques. One would be interviews which would be conducted by interviewing theater staff and managers to understand current problems with manual booking. Another technique would be observation which would be used to watch how ticket sales and reservations are currently handled. Another technique would be user feedback conducted by asking potential users what they expect from an online booking system such as navigation, layouts, and booking confirmations.

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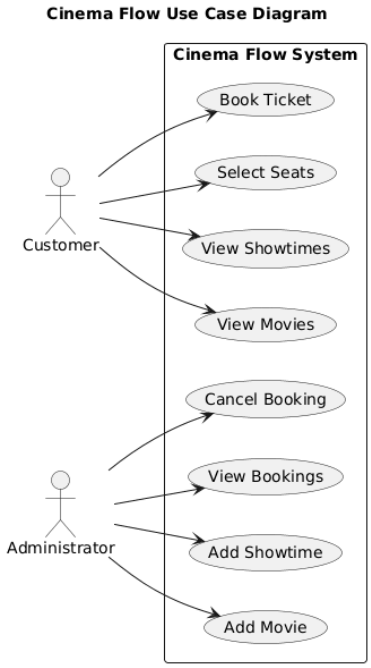
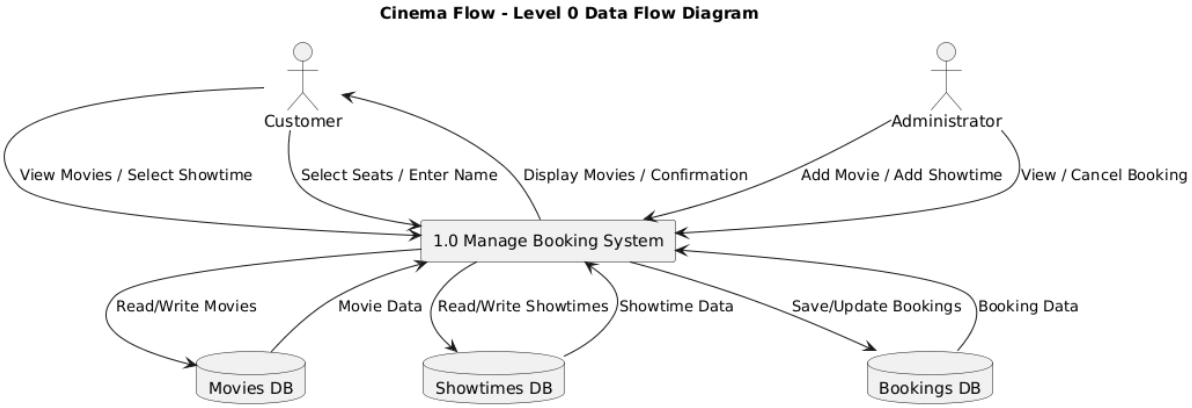
3.4 Use Case Diagram



Part 4 Process and Data Modeling

4.1 Level 0 Data Flow Diagram



4.2 Data Dictionary

Data Element	Description	Type
movie_id	Unique ID for each movie	Integer

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Data Element	Description	Type
title	Name of the movie	String
rating	Movie rating (PG, R, etc.)	String
runtime_minutes	Length of movie	Integer
showtime_id	Unique ID for each showtime	Integer
show_datetime	Date and time of showing	Date Time
price	Ticket price	Decimal
booking_id	Unique booking identifier	Integer
customer_name	Name of customer	String
seat_id	Unique seat identifier	Integer
seat_number	Seat number (e.g., A1, B5)	String
status	Booking status (Active/Cancelled)	String

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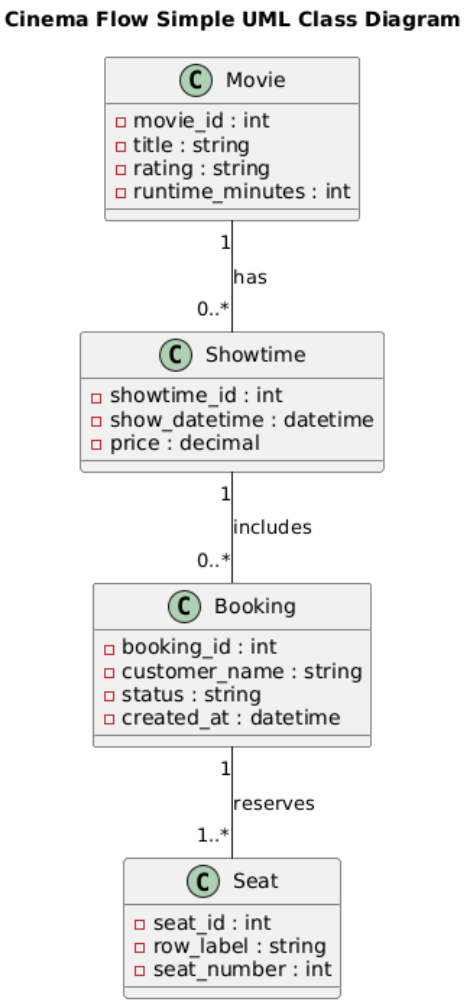
4.3 UML Class Diagram



<u>Part 5 Development Strategy and Interface Concepts</u>

5.1 Development Approach

- The development approach the Cinema Flow system will use is in-house development using a web-based solution. This approach is justified because Cinema Flow is a small movie theater company with specific business needs. By developing the system in-house, the company can customize its features to match its workflow.

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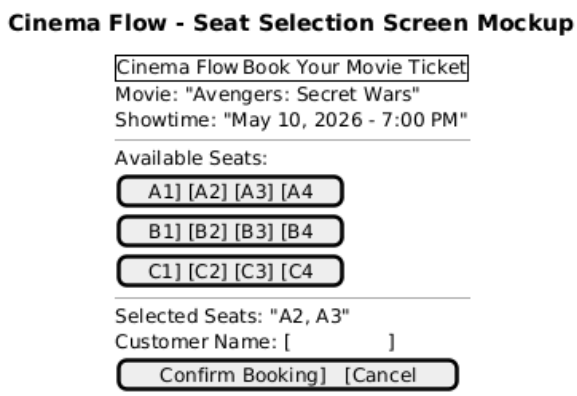
5.2 Basic Interface Design

- **Customer Interface:** The customer side of the system should allow users to view movies currently showing, select showtimes, and view available seats.
- **Administrator Interface:** The administrator side of the system should allow staff to add new movies and showtimes, view all bookings, and cancel bookings when necessary.

5.3 Input Screen Mockup

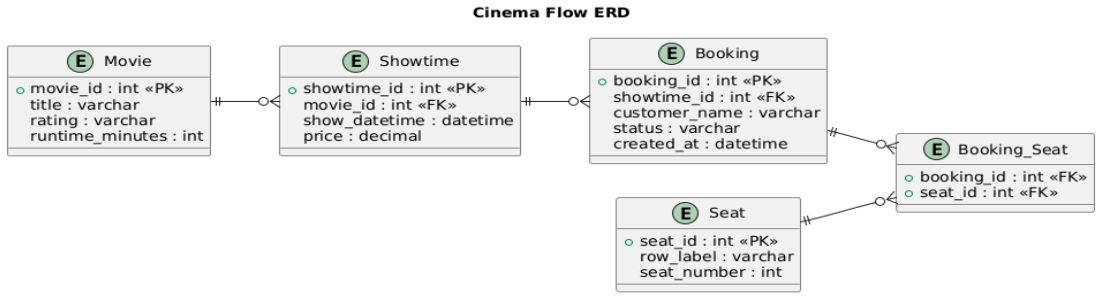


5.4 User-Friendliness

- To ensure the system is user-friendly, one principle that will be applied is simplicity. The screens will avoid clutter and only provide the information needed for the current task. Another principle used will be clear navigation, meaning buttons and links will be clearly labeled so users know what action to take next. Another principle used will be visual feedback, meaning that seat selection should be visually distinguishable between available, selected, and sold seats.

<u>Part 6 Data and Architecture Design</u>

6.1 Entity Relationship Diagram (ERD)



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6.2 System Architecture

- The most appropriate architecture for the Cinema Flow system would be a web-based three-tier architecture.
    - **Presentation Layer:** This is the user interface that customers and administrators interact with through a web browser.
    - **Application Layer:** This layer contains the business logic of the system. For this specific system, this layer handles operations such as displaying available movies, process bookings, and prevent double-booking.
    - **Data Layer:** This layer stores system information in a database. This includes movies, showtimes, bookings and seat data.
- This architecture fits this system because it supports access from multiple devices through a browser. It also separates the interface, application logic, and database for easier maintenance. It improves security by separating user access from direct database access. Finally, this architecture is the best fit because it is well suited for online ticket booking and administrative management.